Design and Implementation of a Microcontroller Based Four-Player Quiz Buzzer System

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ABSTRACT This paper presents the design and implementation of a four-player quiz buzzer system based on the STM32F103C8T6 microcontroller. The system is intended for academic competitions, classroom assessments, and quiz events where rapid and fair participant response detection is required. The developed system incorporates a first-response lockout mechanism, countdown timer, answer timer, visual indication through LEDs and LCD display, buzzer notification, and score management functionality. The implementation ensures that only the first contestant pressing the buzzer gains the opportunity to answer, thereby eliminating ambiguity and improving fairness. Experimental evaluation demonstrates reliable operation with low response latency and user-friendly interaction.

KEYWORDS Quiz Buzzer, STM32F103C8T6, Embedded System, LCD Display, Real-Time Response Detection, Microcontroller Applications.

I. INTRODUCTION

Quiz competitions are widely used in educational institutions to assess knowledge, encourage learning, and promote healthy competition among participants. Conventional quiz systems often rely on manual observation by moderators, which may introduce response ambiguity and human error.

To address this issue, electronic quiz buzzer systems have been developed to accurately identify the first participant who responds. This work presents a microcontroller-based four-player quiz buzzer system that detects the first button press, locks out other participants, provides timing control, and manages scoring automatically.

The proposed system uses an STM32F103C8T6 microcontroller as the central processing unit, interfaced with push buttons, LEDs, LCD display, buzzer, and serial communication.

II. SYSTEM ARCHITECTURE

The complete system consists of an STM32F103C8T6 Microcontroller, four Player Push Buttons, I2C LCD Display, buzzer Module, correct answer Button, wrong Answer Button, UART monitoring interface

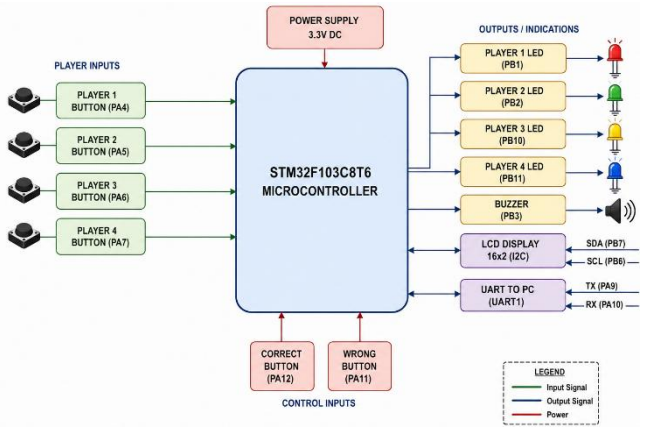


Fig. 3. Operational Flowchart of the Four-Player Quiz Buzzer System

III. HARDWARE DESIGN

The system is built around STM32F103C8T6.Four push buttons are used for players and two buttons are used by the judge for correct and wrong decisions. LEDs indicate the players who pressed first. A buzzer provides audio feedback. An I2C LCD displays messages and scores. UART is used for monitoring.

A. GPIO Pin Mapping

TABLE I
GPIO PIN MAPPING

Function	GPIO Port	Pin
Player 1 Button	PA	4
Player 2 Button	PA	5
Player 3 Button	PA	6
Player 4 Button	PA	7
Wrong Button	PA	11
Correct Button	PA	12
Player 1 LED	PB	1
Player 2 LED	PB	2
Player 3 LED	PB	10
Player 4 LED	PB	11
Buzzer	PB	3
LCD SDA	PB	7
LCD SCL	PB	6
UART TX	PA	9
UART RX	PA	10

B. System Specifications

- Microcontroller: STM32F103C8T6
- Operating Voltage: 3.3 V
- Question Timer: 10 seconds
- Answer Timer: 10 seconds
- Score (Correct): +10
- Score (Wrong): −5
- Communication: UART at 9600 baud
- Display: 16x2 I2C LCD

IV. SOFTWARE DESIGN

The firmware is written using STM32 HAL library. The system works using a state machine with timers.

A. System Specifications

```
uint8_t  system_locked = 0;
uint8_t  winner        = 0;
uint32_t countdown     = 10;
uint8_t  answer_mode   = 0;
uint32_t answerTime    = 10;
int      playerScore[4] = {0,0,0,0};
```

B. System Specifications

```
if(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_4)==GPIO_PIN_RESET
   && system_locked==0)
{
   Winner_Action(1);
}
// Similar code for PA5, PA6, PA7
```

C. Winner Action Function

```
void Winner_Action(uint8_t player)
{
   system_locked = 1;
   winner = player;
   answer_mode = 1;
   answerTime = 10;
}
```

D. Correct Answer

```
playerScore[winner-1] += 10;     // +10 point
```

E. Wrong Answer

```
playerScore[winner-1] -= 5;      // -5 point
```

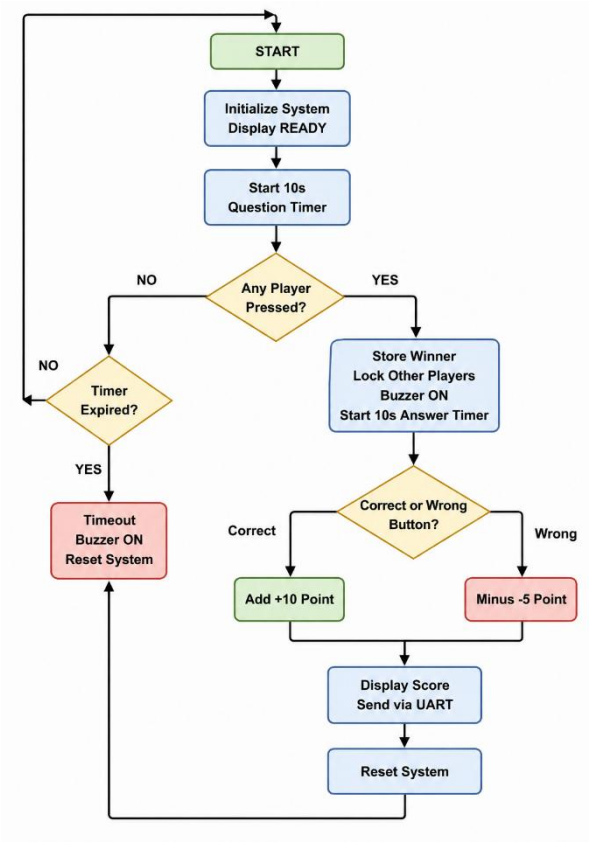
F. System Reset

```
void Reset_System(void)
{
   system_locked = 0;
   winner = 0;
   countdown = 10;
   answer_mode = 0;
}
```

G. UART Configuration

```
huart1.Init.BaudRate    = 9600;
huart1.Init.WordLength  = UART_WORDLENGTH_8B;
huart1.Init.StopBits    = UART_STOPBITS_1;
huart1.Init.Parity      = UART_PARITY_NONE;
huart1.Init.Mode        = UART_MODE_TX_RX;
```

V. FLOWCHART



VI. ALGORITHM

A. Quiz Buzzer Operation

1. Initialize peripherals.
2. Display READY state.
3. Start 10-second question timer.
4. Monitor all player buttons.
5. Detect first valid press.
6. Store winner ID.
7. Lock all other players.
8. Activate buzzer.
9. Start answer timer.
10. Wait for moderator decision.
11. If correct → Score += 10.
12. If wrong → Score -= 5.
13. Update LCD and UART.
14. Reset system.
15. Repeat.

VII. HARDWARE PICTURE

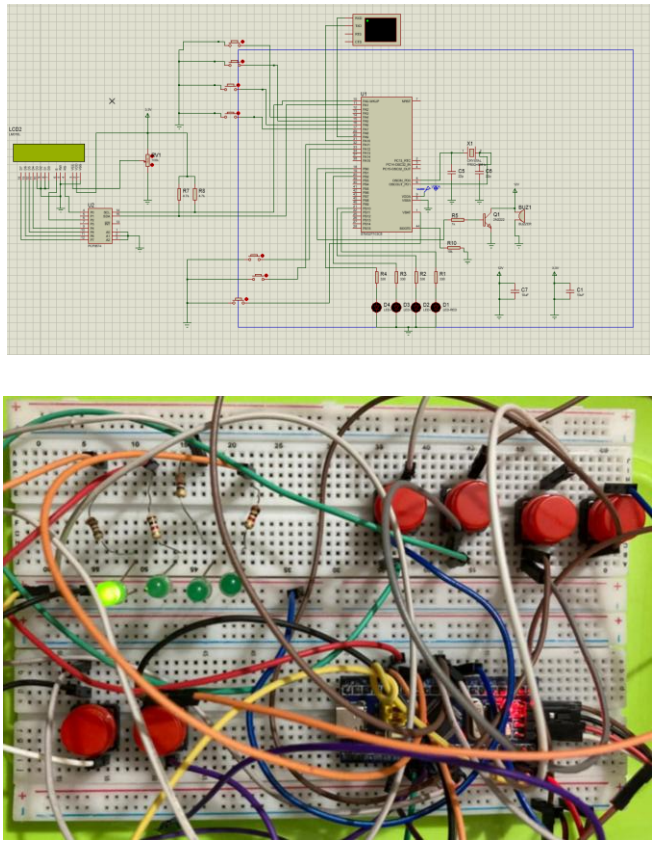


Fig. 4. Proteus Simulation Circuit of the Proposed Quiz Buzzer System

VIII. OUTPUT PICTURE

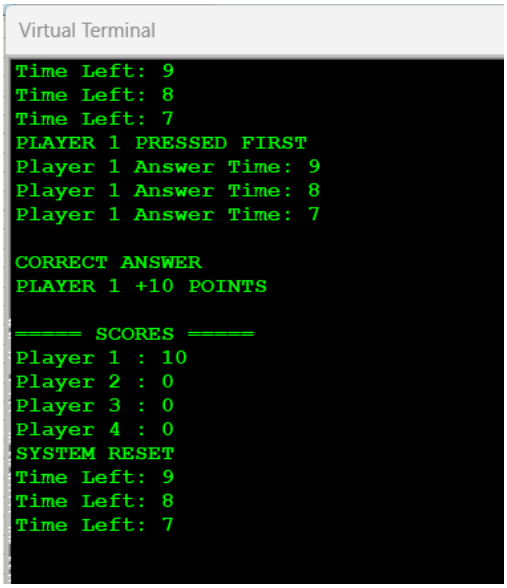


Fig. 5. Experimental Output Showing Winner Detection and Score Update

IX. EXPERIMENTAL RESULT

TABLE II
EXPERIMENTAL RESULTS

Parameter	Result
Number of Players	4
Response Detection	Successful
Lockout Function	Successful
LCD Operation	Successful
UART Monitoring	Successful
Answer Timing	Accurate
Score Update	Successful

X. CONCLUTION

A four-player STM32-based quiz buzzer system has been successfully designed and implemented. The system provides reliable first-response detection, contestant lockout, answer timing, score management, LCD visualization, and UART monitoring. Experimental evaluation confirms the effectiveness of the system in educational and competitive quiz environments. Future work may include wireless buzzers, cloud-based score storage, and mobile application integration.

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